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MAE 283A

MAE 283A Final Project Report

This final project seeks to evaluate data from a microgrid system undergoing a step excitation experiment. The experiment involved a Princeton Power Systems BIGI 250 inverter inside the Kaiser Microgrid. This inverter commanded several step changes in the real power demand by the system in the range of +/- 100 kW. The inverter was connected to a Battery Energy Storage System (BESS) where the dynamic response of the BESS to these step changes was measured as the system output using Phasor Measurement Units. (PMUs) This data set was stored in a MATLAB .mat file called step_Pinv_Pbess.mat, provided by the ChargeBliss company to the MAE 283A course, available on the course website here for download.

Non-parametric data analysis performed on this data included estimates for input auto correlation functions, input spectrum, along with a Bode plot of the SPA estimate G^_0(e^jomega). These analyses indicated that the primary region of excitation for this system is focused near small values of omega between 0.0 rad/s to 2.38745 rad/s. (Figure 1) This revelation in behavior, along with the Bode plot (Figure 2) suggest that downsampling of the system may be useful as it would reduce variance in the spectral estimate and improve researchers abilities to estimate system behaviors.

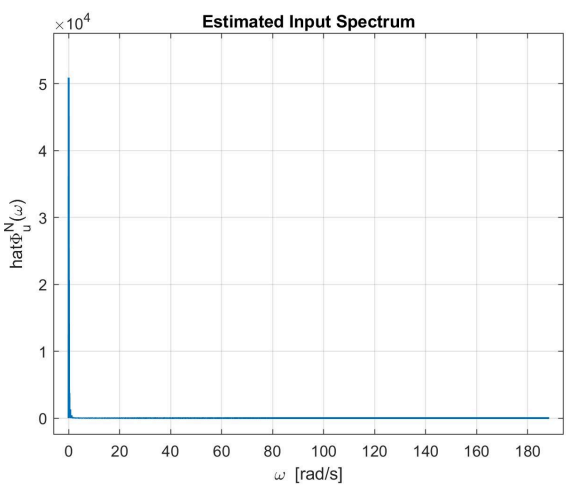

Figure 1: Plot of estimated input spectrum of the system.

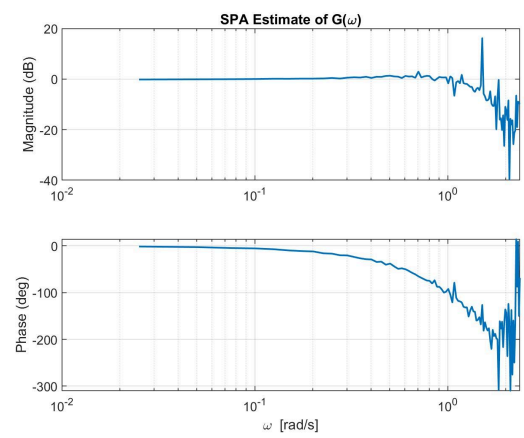

Figure 2: Bode plot with both magnitude and phase of the SPA estimate.

Parametric identification of the system presented some challenges but ultimately resulted in a well fitting Box Jenkins model of the output that the researcher believes to be a good representation of the system. First, early efforts for parametric identification of this system were focused on the generation of an FIR model that could successfully pass model validation tests. Model validation tests used for all parametric models in this section was an estimate of the cross correlation function of the estimate with 99% confidence interval (CI) bounds, both fixed and variable. The FIR system model for early values of

tau suggests that the model would be able to pass validation, but it ultimately exits the CI and therefore cannot be validated. (Figure 3) Realization methods based in part on this same FIR model, were used to try and compute a more accurate model for this system. The Realization model was verified again with a cross correlation function between the input u(t) and the simulation error of the system. (Figure 4) The realization model did a better job and the cross correlation function remained within the bounds of the 99% CI. The values are frighteningly close to exiting the CI, but they do remain inside, suggesting that this new state space model does pass the model validation test.

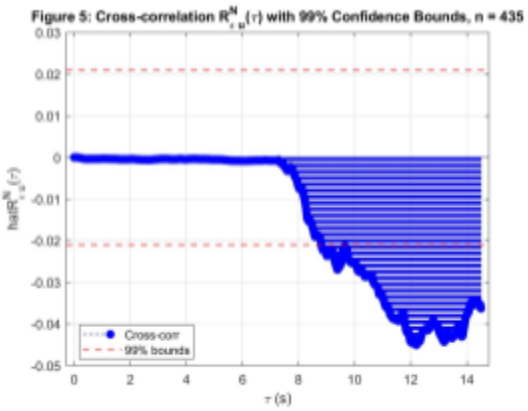

Figure 3: Cross Correlation for FIR model

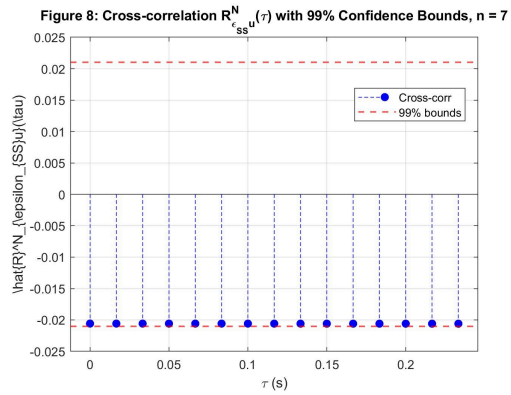

Figure 4: Cross Correlation for Realization model

The final, and best fitting model tested was a Box Jenkins(BJ) model of the 3rd order. Through testing of various orders of BJ, this model was determined to do the best job at staying within the CI. In further testing of the system, the simulated output and predicted output both were able to follow the true systems output and transient behaviors with high levels of accuracy.(Figures 5 and 6) This model vastly improved upon all other models mentioned thus far in this report, showcasing that despite the high levels of complexity of a box jenkins style parametric model, the model was able to properly capture the behavior of the step excitation experiment ran at the Kaiser Microgrid.

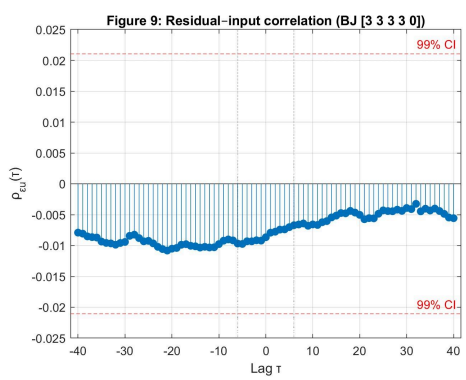

Figure 5: Cross Correlation for BJ model

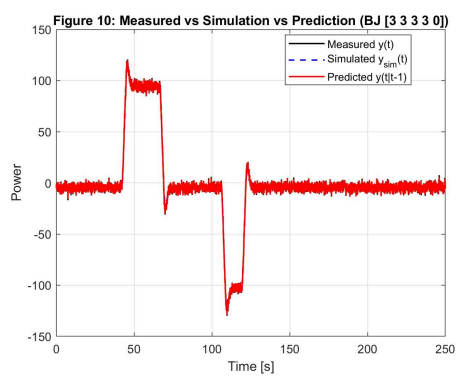

Figure 6: Measured vs simulation vs prediction for BJ model